

Reference excerpt 01-010

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due to the internal jugular vein (IJV) pulsation[Mackenzie(1902)]. The pressure changes generated in the RA are indeed transmitted to the IJV and affect the velocity of the blood in two ways i) modifying the pressure gradient so that it is no longer constant over time and generates a non-steady flow [Sisini et al.(2016), Kalmanson et al. (1972)], ii) cyclically varying the CSA of the IJV [Sisini et al.(2015)]. Since the walls of the IJVs are neither rigid nor collapsed, the pressure variations generated in the RA are transmitted to the IJV as a pressure wave with finite 1 Figure 1: Geometrical model of the jugular-heart segment. The z axis is positive in the jugular-heart direction. l is the distance between the RA and point G in which the scan takes place. Blood velocity is indicated by w . propagation velocity c . Such pressure waves are responsible for the modulated component of the velocity of blood in the IJV. For this reason, their wave equation can be used to derive the instantaneous velocity of the blood even in the absence of a direct measurement of such parameter obtained, for example, by using an ultrasound Doppler scanner. A first attempt to this goal was presented in [Sisini et al.(2015b)], where the instantaneous velocity of the blood in the IJV was determined qualitatively, for one cardiac cycle, using the Womersley equation [Womersley(1955a), Womersley(1955b), Womersley(1955c)]. Physical description of the RA-IJV segment The RA-IJV system is represented in Fig. 1 as a tube with a circular section. G